

$$\text{PAPER_1555} - \text{Z Boson Mass} = m_Z = N_{CH} \cdot SO_5 + F_{TRZ} \cdot SO_5 + F_{TRZ}^2 \cdot (SO_5 + D_{BSFG} + D_{phys} + SSQ - SSQ^3) \approx 91.20 \text{ GeV}$$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Particle Physics **Date:** June 18, 2026 **Status:** CLOSED — 0.018% closure from PAPER_1209HH_S654

Observation

PAPER_1209HH_S654 (Particle Physics Unified Proof Set) lists **Z Boson Mass** with the integer-primitive expression:

$$Z \text{ Boson Mass} = m_Z = N_{CH} \cdot SO_5 + F_{TRZ} \cdot SO_5 + F_{TRZ}^2 \cdot (SO_5 + D_{BSFG} + D_{phys} + SSQ - SSQ^3) \approx 91.20 \text{ GeV}$$

Status: **0.018%**.

UQFF Closed Identity

$$m_Z = N_{CH} \cdot SO_5 + F_{TRZ} \cdot SO_5 + F_{TRZ}^2 \cdot (SO_5 + D_{BSFG} + D_{phys} + SSQ - SSQ^3) \approx 91.20 \text{ GeV} \quad 0.018\%$$

Physical Interpretation

Dominant: $N_{CH} \cdot SO_5 = 9 \cdot 10 = 90 \text{ GeV}$ — sits at the Z mass minus 1 GeV. Small corrections bring it to 91.20 vs CODATA 91.1876.

NOT REPLACEMENT

CODATA/PDG treats Z Boson Mass as a precision-measured constant. UQFF supplies the structural integer-primitive form, demonstrating the framework's predictive economy without claiming to replace experimental measurement.

Reference

- Source: PAPER_1209HH_S654
 - Related: PAPER_1209 series unified proof sets
 - Calculator dispatch: `calculate_paradox({"paradox": "m_z_91_188"})`
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